Kalman Filter for Yield Curve Prediction

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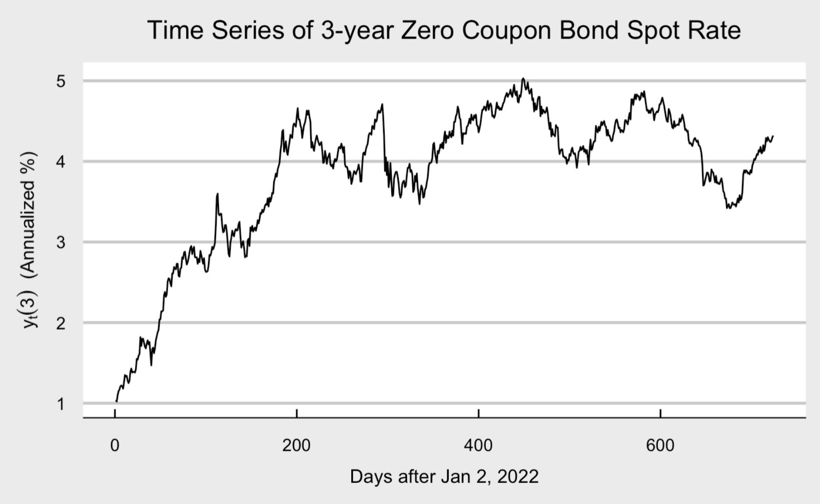
Content Structure

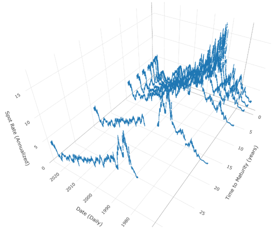
Problem Statement

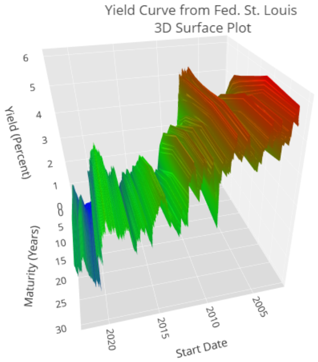
We are familiar with univariate time series modeling.



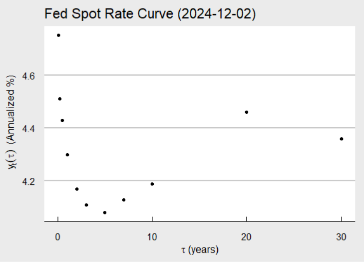
Problem Statement

We are interested in modeling the structure of the yield curve at different times to maturity:

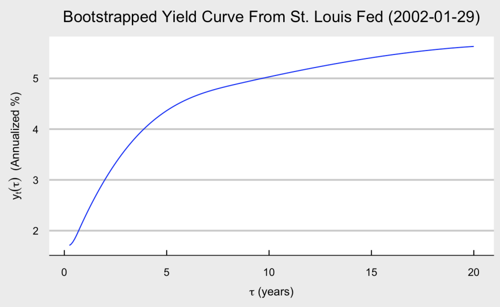




Visualizing the Data Set



Single Day



Full Data Set

Dynamic Nelson Siegel Model

Dynamic Nelson Siegel Model (DNS):

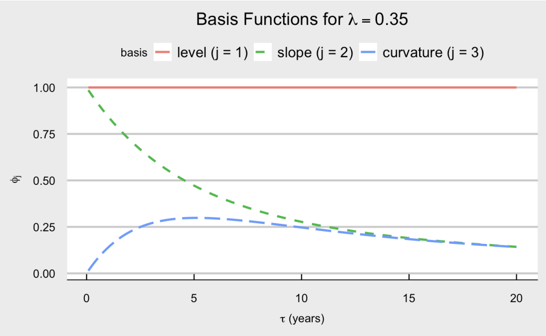
$$Y_t(\tau_i) = L_t + S_t \left(\frac{1 - e^{-\lambda \tau_i}}{\lambda \tau_i} \right) + C_t \left(\frac{1 - e^{-\lambda \tau_i}}{\lambda \tau_i} - e^{-\lambda \tau_i} \right) + \epsilon_t(\tau_i)$$

In Matrix Form:

$$\mathbf{Y}_t(\boldsymbol{\tau}) = \boldsymbol{\Phi}(\lambda, \boldsymbol{\tau}) \boldsymbol{\beta}_t + \boldsymbol{\epsilon}_t$$

- $\boldsymbol{\tau} \in \mathbb{R}^N$: times to maturity
- $\mathbf{Y}_t(\boldsymbol{\tau}) \in \mathbb{R}^N$: yields to maturity in matrix form
- $\lambda \in \mathbb{R}^+$: decay factor
- $\boldsymbol{\beta}_t = [L_t, S_t, C_t]^{\mathrm{T}} \in \mathbb{R}^3$: Level, Slope, Curvature
- $\boldsymbol{\Phi}(\lambda, \boldsymbol{\tau}) \in \mathbb{R}^{N \times 3}$: matrix of basis functions
- $\boldsymbol{\epsilon}_t \overset{i.i.d}{\sim} MVN(0, \Sigma)$: error term Σ can be assumed to be diagonal $\Sigma = \sigma^2 \cdot \mathbb{I}$ or not

With $\lambda = 0.35$:



Basis Functions

- $\phi_1 = 1$
- $\phi_2 = \frac{1-e^{-\lambda\tau}}{\lambda\tau}$
- $\phi_3 = \frac{1-e^{\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau}$

Frame Title

$$
\begin{bmatrix} y_i(\tau_1) \\ y_i(\tau_2) \\ \vdots \\ y_i(\tau_n) \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \frac{1-e^{-\lambda\tau_1}}{\lambda\tau_1} & \frac{1-e^{-\lambda\tau_1}}{\lambda\tau_1} - e^{-\lambda\tau_1} \\ 1 & \frac{1-e^{-\lambda\tau_2}}{\lambda\tau_2} & \frac{1-e^{-\lambda\tau_2}}{\lambda\tau_2} - e^{-\lambda\tau_2} \\ \vdots & \vdots & \vdots \\ 1 & \frac{1-e^{-\lambda\tau_n}}{\lambda\tau_n} & \frac{1-e^{-\lambda\tau_n}}{\lambda\tau_n} - e^{-\lambda\tau_n} \end{bmatrix}}_{\Phi} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}.
$$

Level(β_0): Represents the long-term interest rate
Slope(β_1): Represents the short-term vs. long-term rate differential
Curvature(β_2): Measures the medium-term hump in the yield curve

OLS on Nelson Siegel Model

Assumption: $\boldsymbol{\Sigma} = \sigma^2 \cdot \mathbb{I}$, where $\mathbb{I}$ is the identity matrix
Using OLS[1] algorithm, the estimator $\hat{\boldsymbol{\beta}}$ obtained by maximizing likelihood is:

$$\hat{\boldsymbol{\beta}} = \frac{1}{\mathrm{T}} \sum_{t=1}^{\mathrm{T}} \left(\boldsymbol{\Phi}(\lambda; \boldsymbol{\tau})^\top \boldsymbol{\Phi}(\lambda; \boldsymbol{\tau}) \right)^{-1} \boldsymbol{\Phi}(\lambda; \boldsymbol{\tau})^\top \mathbf{Y}_t(\boldsymbol{\tau})$$

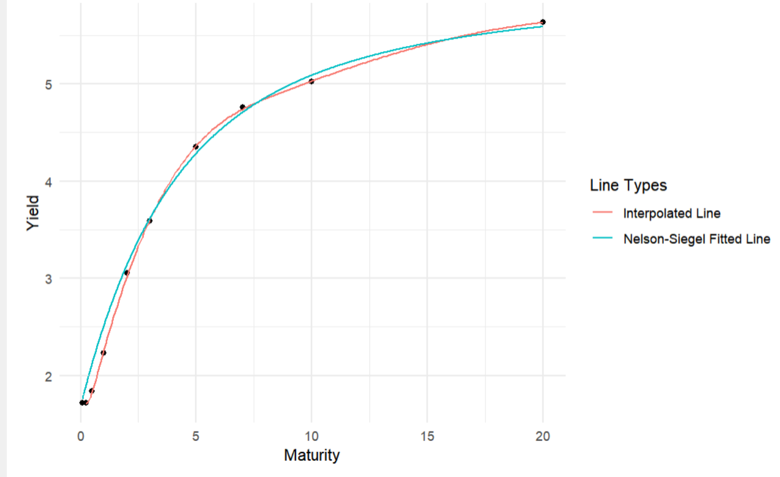
$$\begin{bmatrix} y_i(\tau_1) \\ y_i(\tau_2) \\ \vdots \\ y_i(\tau_n) \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \frac{1-e^{-\lambda\tau_1}}{\lambda\tau_1} & \frac{1-e^{-\lambda\tau_1}}{\lambda\tau_1} - e^{-\lambda\tau_1} \\ 1 & \frac{1-e^{-\lambda\tau_2}}{\lambda\tau_2} & \frac{1-e^{-\lambda\tau_2}}{\lambda\tau_2} - e^{-\lambda\tau_2} \\ \vdots & \vdots & \vdots \\ 1 & \frac{1-e^{-\lambda\tau_n}}{\lambda\tau_n} & \frac{1-e^{-\lambda\tau_n}}{\lambda\tau_n} - e^{-\lambda\tau_n} \end{bmatrix}}_{\boldsymbol{\Phi}} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}.$$

The corresponding unbiased estimator $\hat{\sigma}^2$ is:

$$\hat{\sigma}^2 = \frac{1}{N\mathrm{T} - 3} \sum_{t=1}^{\mathrm{T}} \left(\mathbf{Y_t}(\boldsymbol{\tau}) - \boldsymbol{\Phi}(\lambda; \boldsymbol{\tau})\hat{\boldsymbol{\beta}} \right)^\top \left(\mathbf{Y_t}(\boldsymbol{\tau}) - \boldsymbol{\Phi}(\lambda; \boldsymbol{\tau})\hat{\boldsymbol{\beta}} \right)$$

[1] **legendre1805comets**

Advantages



Interpretability: Three predictors has its econometric interpretations compared with other regression techniques

Dimensionality: NS model uses only a 3 parameters.

State Space Model for Nelson Siegel Model

$$\mathbf{X}_t = \mathbf{A}\,\mathbf{X}_{t-1} + \mathbf{B}\,\mathbf{W}_t \quad \mathbf{W}_t \sim \mathcal{MVN}(\mathbf{0}, \mathbf{Q}), \tag{1}$$

$$\mathbf{Y}_t = \mathbf{C}\,\mathbf{X}_t + \mathbf{D}\,\mathbf{Z}_t, \quad \mathbf{Z}_t \sim \mathcal{MVN}(\mathbf{0}, \mathbf{R}) \tag{2}$$

- $\mathbf{X}_t \in \mathbb{R}^3$: The latent state at time t.
- $\mathbf{A} \in \mathbb{R}^{3\times 3}$: The state transition matrix.
- $\mathbf{B} \in \mathbb{R}^{3\times p}$: State noise transformation matrix
- $\mathbf{W}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$: The process noise.
- $\mathbf{Y}_t \in \mathbb{R}^n$: The measurement vector at time t.
- $\mathbf{C} \in \mathbb{R}^{n\times 3}$: The observation matrix.
- $\mathbf{D} \in \mathbb{R}^{n\times r}$: Observation noise transformation matrix
- $\mathbf{Z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$: The measurement noise.

State Space Model for NS GARCH(1,1)

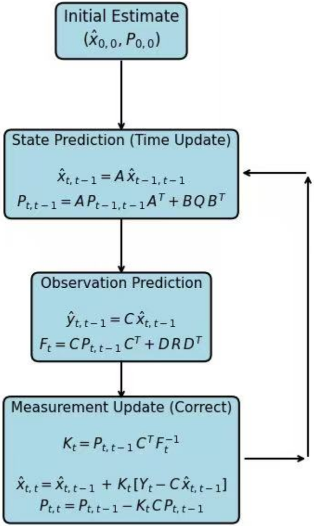
$$\mathbf{X}_t = \mathbf{A}\,\mathbf{X}_{t-1} + \mathbf{B}\,\mathbf{W}_t \quad \mathbf{W}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q_t}), \tag{3}$$

$$\mathbf{Y}_t = \mathbf{C}\,\mathbf{X}_t + \mathbf{D}\,\mathbf{Z}_t, \quad \mathbf{Z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}) \tag{4}$$

$$\mathbf{Q}_t = \boldsymbol{\omega}_0 + \boldsymbol{\alpha}_1 \mathbf{W}_{t-1}^2 + \boldsymbol{\beta}_1 \mathbf{Q}_{t-1}. \tag{5}$$

- $\mathbf{Q}_t \in \mathbb{R}^{3\times3}$: The time-varying process noise covariance matrix.
- $\boldsymbol{\omega}_0 \in \mathbb{R}^{3\times3}$: The baseline (constant) covariance term.
- $\boldsymbol{\alpha}_1 \in \mathbb{R}^{3\times3}$: The coefficient controlling the influence of past process noise on current volatility.
- $\mathbf{W}_{t-1} \in \mathbb{R}^{3\times3}$: The process noise at time t, which follows $\mathbf{W}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t)$.
- $\boldsymbol{\beta}_1 \in \mathbb{R}^{3\times3}$: The persistence coefficient determining how previous covariance $\mathbf{Q}_{t-1}$ affects the current covariance $\mathbf{Q}_t$.

Visualizing Kalman Filter



Recursion via the Marginal Likelihood.

- The *log-marginal-likelihood* of the observations:

$$\ell(\Theta) \;=\; \log p(\mathbf{Y}_{1:N} \mid \Theta) \;=\; \sum_{t=1}^{N} \log p(\mathbf{Y}_t \mid \mathbf{Y}_{1:t-1}, \Theta).$$

- Each term $\log p(\mathbf{Y}_t \mid \mathbf{Y}_{1:t-1}, \Theta)$ can be written in terms of

$$e_t \;=\; \mathbf{Y}_t - \mathbf{C}\,\hat{x}_{t|t-1}, \quad F_t \;=\; \mathbf{R} \;+\; \mathbf{C}\,\mathbf{P}_{t|t-1}\,\mathbf{C}^T,$$

which come from the Kalman filter's prediction step for each t.

Parameter Estimation for the State Transition Matrix $\mathbf{A}_t$

The state transition matrix $\mathbf{A}_t$ is modeled as a time-varying matrix using Yule-Walker Estimate for an **AR(1) process**:

$$\mathbf{A}_t = \begin{bmatrix} a_{LL,t} & 0 & 0 \\ 0 & a_{SS,t} & 0 \\ 0 & 0 & a_{CC,t} \end{bmatrix}$$

Each diagonal element follows an **AR(1) process**:

$$a_{ii,t} = \rho_i a_{ii,t-1} + \gamma_i + \epsilon_{i,t}, \quad i \in \{L, S, C\}$$

where:

- $a_{ii,t}$: The state transition coefficient for the i-th Nelson-Siegel parameter at time t.
- ρ_i: The autoregressive coefficient .
- γ_i: The drift term capturing the long-run mean.
- $\epsilon_{i,t} \sim \mathcal{N}(0, \sigma_i^2)$: White noise innovation term.

Parameter Estimation for $\mathbf{Q}_t$

We model the time-varying covariance matrix $\mathbf{Q}_t$ for the Nelson-Siegel parameters $X_t = [L_t, S_t, C_t]^T$ as a **diagonal matrix**:

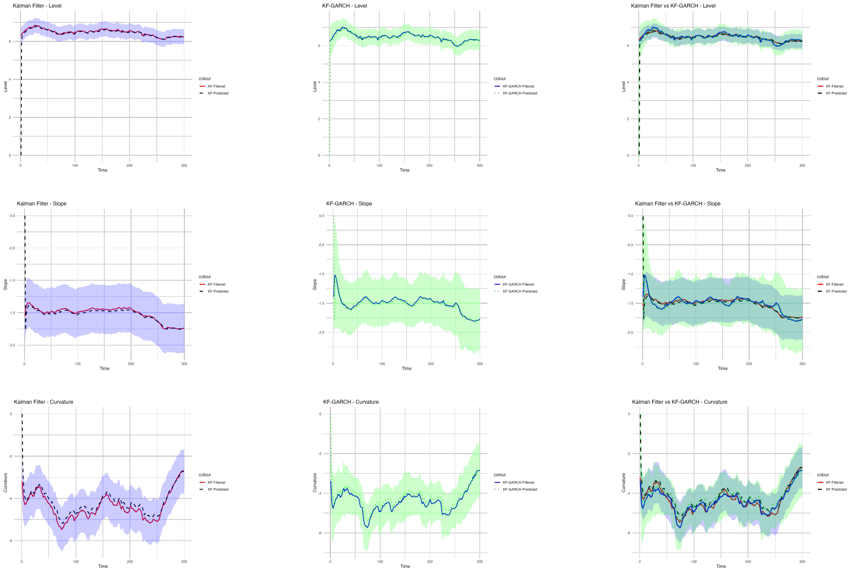
$$\mathbf{Q}_t = \begin{bmatrix} q_{LL,t} & 0 & 0 \\ 0 & q_{SS,t} & 0 \\ 0 & 0 & q_{CC,t} \end{bmatrix}$$

Each diagonal element follows a **GARCH(1,1) process**:
Expanded Form:

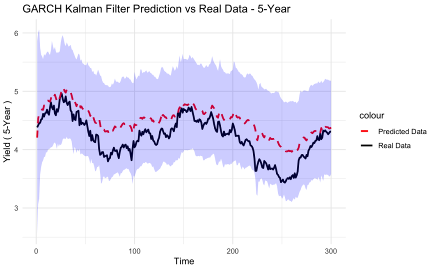
$$\begin{aligned} q_{LL,t} &= \alpha_0^{(L)} + \alpha_1^{(L)} w_{L,t-1}^2 + \beta_1^{(L)} q_{LL,t-1}, \\ q_{SS,t} &= \alpha_0^{(S)} + \alpha_1^{(S)} w_{S,t-1}^2 + \beta_1^{(S)} q_{SS,t-1}, \\ q_{CC,t} &= \alpha_0^{(C)} + \alpha_1^{(C)} w_{C,t-1}^2 + \beta_1^{(C)} q_{CC,t-1}. \end{aligned}$$

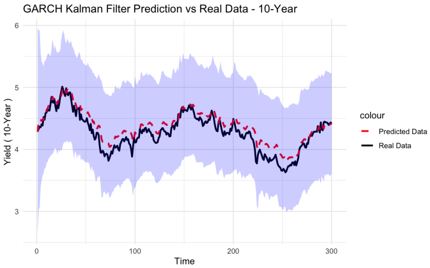
Since $\mathbf{Q}_t$ is diagonal, the off-diagonal elements are assumed to be zero, meaning the Nelson-Siegel parameters have independent conditional variances.

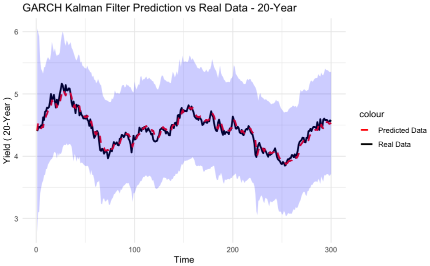
Comparison of KF and KF-GARCH Predictions

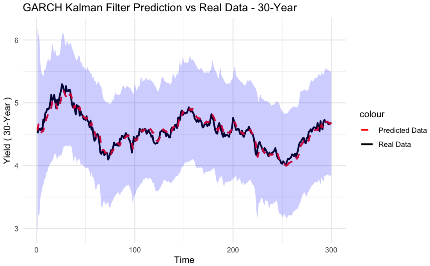


Kalman Filter with Volatility









Kalman Filter w/o volatility

Future Studies

- Implementation of gradient descent for parameter estimation.
- Compare results with other optimization methods (e.g., EM algorithm).
- Apply particle filter for non-Gaussian Noise assumption in Bond data.